

Midterm solution

1.

(a) We adopt inverse transform method.

$$F(x) = \int_0^x \frac{2}{(1+t)^2} dt = 2 \left(1 - \frac{1}{1+x} \right) = \frac{2x}{1+x}.$$

Let $U \sim \text{Unif}(0, 1)$. Solving $u = \frac{2x}{1+x}$ gives $x = \frac{u}{2-u}$. Hence

$$X = \frac{U}{2-U}.$$

(b) Take $g(x) = 1$ on $[0, 1]$. Then $\max_x f(x)/g(x) = 2$, so choose $C = 2$. Algorithm:

- Draw $Y, V \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$;
- Accept Y if $V \leq \frac{f(Y)}{Cg(Y)} = \frac{1}{(1+Y)^2}$; Otherwise, reject and repeat.

(c) If $u = 0.71$, then $x = u/(2-u) = 0.71/1.29 \approx 0.55$.

(d) Here

$$\frac{f(x)}{g(x)} = \frac{2/(1+x)^2}{2x} = \frac{1}{x(1+x)^2} \rightarrow \infty \text{ as } x \rightarrow 0,$$

so no finite C exists and the A-R algorithm is invalid.

2. Notice

$$P(W = i) = P(X = i | Y = j) = \frac{P(Y = j | X = i) P(X = i)}{P(Y = j)}.$$

Hence

$$\frac{P(W = i)}{P(X = i)} = \frac{P(Y = j | X = i)}{P(Y = j)} \leq \frac{1}{P(Y = j)}.$$

Take $C = \frac{1}{P(Y=j)}$, and by using the A-R algorithm without knowing normalization constant we can get rid of this dependence. Then the A-R acceptance probability is

$$\frac{P(W = i)}{CP(X = i)} = \frac{P(Y = j | X = i) \frac{P(X=i)}{P(Y=j)}}{\frac{1}{P(Y=j)} P(X = i)} = P(Y = j | X = i).$$

Therefore, $P(Y = j | X = i)$ is the acceptance probability.

3.

(a) There are at least two methods to solve this problem, we show both below.

(i) Composition method. Set $J = 1, 2, 3$ with probability $\frac{1}{3}$. Then, set Y_1 to have CDF x , Y_2 to have CDF x^3 , and Y_3 to have CDF x^5 , for $0 \leq x \leq 1$.

To implement the algorithm, we:

- Generate $U \sim \text{Unif}(0, 1)$.
- If $U \leq \frac{1}{3}$, set $J = 1$; else if $U \leq \frac{2}{3}$, set $J = 2$; else set $J = 3$.
- Generate $V \sim \text{Unif}(0, 1)$
- If $J = 1$, output V ; if $J = 2$, output $V^{\frac{1}{3}}$, if $J = 3$, output $V^{\frac{1}{5}}$.

(ii) **Acceptance-rejection** Density:

$$f(x) = F'(x) = \frac{1 + 3x^2 + 5x^4}{3}, \quad 0 \leq x \leq 1.$$

Using $g(x) = 1$ gives $\max_x f(x)/1 = f(1) = 3$. Algorithm:

- (a) Draw $Y, V \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$;
 - (b) Accept Y if $V \leq f(Y)/3 = \frac{1+3Y^2+5Y^4}{9}$;
 - (c) Otherwise repeat.
- (b) Using composition method. Set Y_t to have $P(Y_t > x) = \frac{1}{(1+2x)^t}$ for $x > 0$. Set T to be $\text{Exp}(1)$.

To implement ITM for Y_t , set $U = 1 - \frac{1}{(1+2x)^t}$. So $x = \frac{1}{2} \left(\left(\frac{1}{1-U} \right)^{\frac{1}{t}} - 1 \right)$. So overall:

- Generate $U \sim \text{Unif}(0, 1)$
- Set $T = -\log(1 - U)$
- Generate $V \sim \text{Unif}(0, 1)$
- Output $\frac{1}{2} \left(\left(\frac{1}{1-V} \right)^{\frac{1}{T}} - 1 \right)$

4. We generate $X \sim \frac{|\cos \frac{1}{x}|}{K\sqrt{x}}$, $0 < x < 1$, for some constant $K > 0$ that makes $\frac{|\cos \frac{1}{x}|}{K\sqrt{x}}$ a density on $0 < x < 1$. We can do this by A-R.

We first propose $g(x) = \frac{1}{2\sqrt{x}}$, which can be verified to be a density on $0 < x < 1$ since $\int_0^1 \frac{1}{2\sqrt{x}} dx = \sqrt{x}|_0^1 = 1$. This density can be simulated by ITM. In fact, since the CDF of X is $\sqrt{x}$, ITM outputs U^2 where $U \sim \text{Unif}(0, 1)$. With the above, we have $\frac{|\cos \frac{1}{x}|}{\frac{1}{2\sqrt{x}}} \leq 2$. So we set $C = 2$ and this allows us to run the A-R algorithm for X . Then, given X , we generate Y from $\text{Unif}\left(0, \frac{|\cos \frac{1}{X}|}{\sqrt{X}}\right)$. That is, the overall algorithm is:

- Generate $U \sim \text{Unif}(0, 1)$ and set $W = U^2$
- Generate $V \sim \text{Unif}(0, 1)$ independent of U
- If $V \leq \frac{f(W)}{Cg(W)} = \frac{|\cos \frac{1}{W}|}{2\frac{1}{\sqrt{W}}} = |\cos \frac{1}{W}|$, then output $X = W$; otherwise, repeat the steps.
- Generate $Z \sim \text{Unif}(0, 1)$.
- Given X , set $Y = \frac{|\cos \frac{1}{X}|}{\sqrt{X}} \cdot Z$.